AOU
الجامعة العربية المفتوحة
Arab Open University

Faculty of Business Studies

Course Code: FIN302-F

Course Title: Portfolio Theory

Mid-term Examination / Version: C

2025-2026 / Fall

4-December-2025

AOU EGYPT

Number of Exam Pages: 3 (including this cover sheet)	Time Allowed: 1.5 hours

Instructions:

1- Take **FIVE (5)** minutes to read all instructions carefully before attempting to write.
2- Mobile phones must be in "SWITCHED **OFF**" mode throughout the examination.
3- This is a **CLOSED** book examination. NO documents of any kind are allowed.
4- You **MUST** answer the questions using only the answer booklets provided
5- You **MUST** return the examination paper together with your answer booklets.
6- Do **NOT** write your answers in pencil except graphs/sketches
7- Students are **NOT** allowed to use any type of dictionary or mobile phones during the examination
8- Students are allowed to use ***non-programmable*** calculators only (if needed).
9- Any form of cheating will **NOT** be tolerated and will subject you to AOU Cheating Policy.

CamScanner

Question 1: Choose the right answer for each of the following questions (25 marks; 5 marks per question)

1. Which of the following is NOT traded in the securities markets?

A) stocks
B) bonds
C) derivatives
D) real estate

2. Which of the following investments represents partial ownership of a corporation?

A) bonds
B) mutual funds
C) commercial paper
D) common stock

3. Use the following tax rates to answer the question:

Tax rates	Individual Returns	Joint Returns
10%	$0 to $9,525	$0 to $19,050
12%	$9,526 to $38,700	$19,051 to $77,400
22%	$38,701 to $82,500	$77,401 to $165,000
24%	$82,501 to $157,500	$165,001 to $315,000
32%	$157,501 to $200,000	$315,001 to $400,000
35%	$200,001 to $500,000	$400,001 to $600,000
37%	Over $500,000	Over $600,000

	Taxable Income	
Tax Rates	Individual Returns	Joint Returns
0%	$0 to $38,600	$0 to $77,200
15%	$38,601 to $425,800	$77,201 to $479,000
20%	Over $425,800	Over $479,000

Jon earned $82,500 in taxable income, all from wages and interest, and files an individual tax return. What is the amount of Josh's taxes for the year 2018? Round to the nearest dollar.

A) $13,750
B) $14,090
C) $18,150
D) $12,285

4. Funds that merely track a broad index and make no attempt to identify undervalued or exceptional growth stocks are known as

A) hedge funds.
B) passively managed funds.
C) actively managed funds.
D) equity funds.

5. Over the long term, which one of the following has historically had the lowest risk and lowest average annual rate of return?

A) U.S. common stock
B) long-term government bonds
C) short-term government bills
D) German common stock

FUE EGYPT

CamScanner

Question 2: Answer the following questions (25 marks):

Describe the steps involved in the investment decision process. Be sure to mention how returns and risks can be evaluated together to determine acceptable investments (25 marks; 6.25 marks for each step).

Question 3: Answer the following questions (25 marks):

A. Given a real rate of interest of 2%, an expected inflation premium of 3%, and risk premiums for investments A and B of 4% and 6%, respectively, find the following:
   1. The risk-free rate of return, r_f (7 marks).
   2. The required returns for investments A and B (8 marks; 4 marks for each investment).

B. Over the past 4 years, the annual rates of return on stock of Brown & Warren Inc. have been 9%, 10%, 7% and -3%, respectively, over the past four years. Compute the standard deviation of these returns (10 marks).

Question 4: Answer the following questions (25 marks):

Returns on the stock of First Boston and Midas Metals for the years 2017-2020 are shown below.

	First Boston	Midas Metals
2017	2.00%	-10.00%
2018	5.00%	-12.00%
2019	-15.00%	4.00%
2020	3.00%	25.00%

a. Compute the average annual return for each stock and a portfolio consisting of 50% First Boston and 50% Midas (9 marks; 3 marks for each average computed).
b. Compute the standard deviation for each stock and the portfolio (9 marks; 3 marks for each St. Dev. computed).
c. Are the stocks positively or negatively correlated and what is the effect on risk? (7 marks).

"End of Exam"

AOU EGYPT

CamScanner